

Course “Software Verification and Certification”

Course Code: CS 448/648

Term: *Spring 2026*

Lecture#3: 13 Jan 2026

**“Systems: Software-Intensive to Interaction-Intensive
to Network-Intensive and a Hybrid Perspective”**

Dr. Gourinath Banda

Professor and Lead_PI - ACPS Lab

Department of Computer Science and Engineering

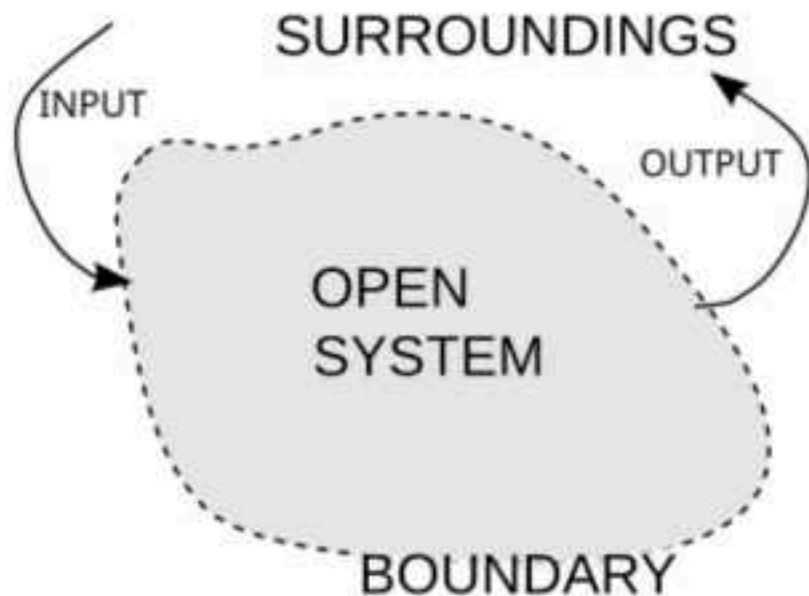
Indian Institute of Technology Indore

Venue @ L-01, LHC

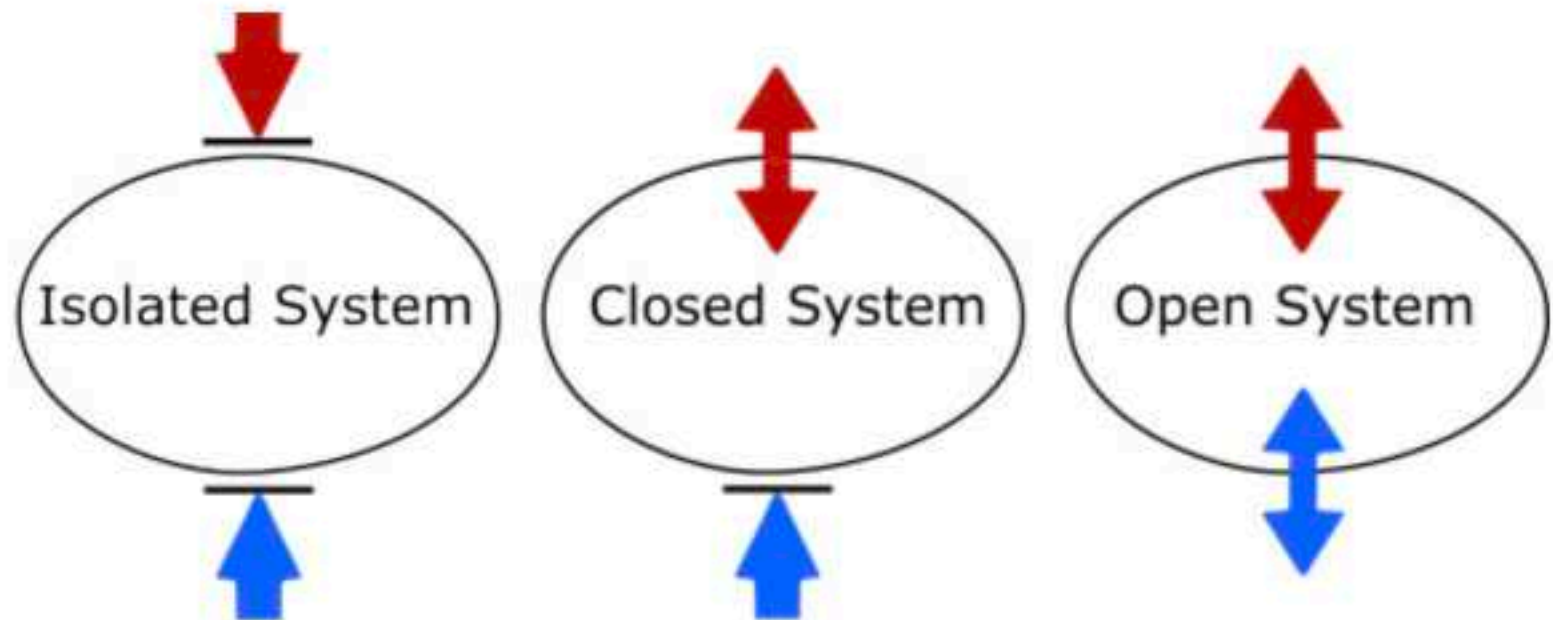
Systems: Basics

- What is a **System**? (Foundation Concept)
 - System is a collection of interacting components/modules/sub-systems, organized to achieve a defined purpose, operating within a boundary, and interacting with an environment.
- **Key characteristics of any system:**
 - **Boundary:** what is inside vs outside the system
 - **Environment:** external entities influencing the system
 - **Interactions:** communication or control among internal components and across the boundary
 - **Components/Modules/Subsystems:** parts with individual responsibilities

Systems: Basics



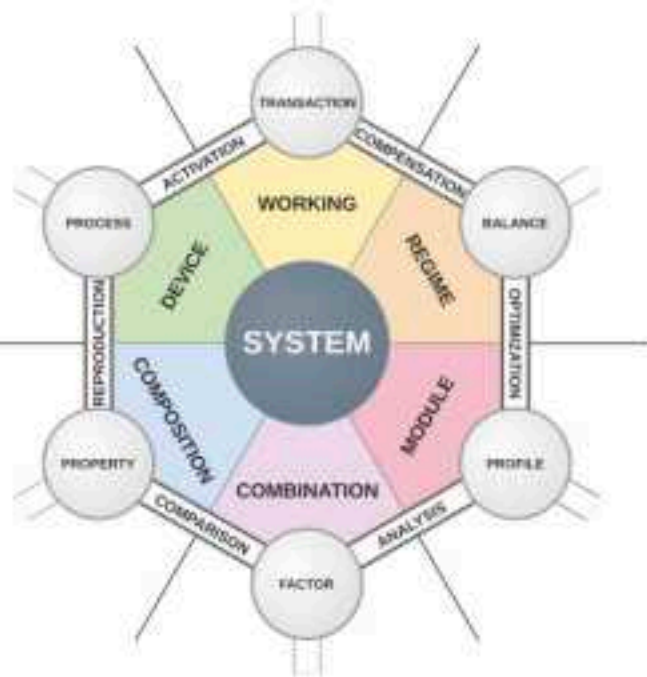
Systems: Basics



 Exchange of energy
 Exchange of matter

Systems: Basics

A System “is **not just a collection** but it also exhibits **emergent behavior**, properties that do not exist in individual components.”



What is a Software System?

- A software system is a system where:
 - Core functionality is implemented in software
 - Behaviour is delivered through program-logic
 - Interactions are information-centric, NOT physical!
- Examples:
 - Operating systems
 - Compilers
 - Databases
 - Web services

Software System: From Verification Perspective

- Software systems typically:
 - Operate in a logical (digital) environment
 - Interact through APIs, files, networks
 - Fail in logical ways (deadlock, race conditions, data corruption)
- At this level, verification focuses on:
 - Functional correctness
 - On the other end of spectrum is non-functional properties
 - Such as temporal properties
 - Safety properties (e.g., no invalid states)
 - Liveness (e.g., requests eventually served)

Software Intensive Systems: Beyond Standalone Software

- A software-intensive system is one where:
 - Software is the dominant source of complexity
 - But the (greater overall) system includes:
 - people, processes, data, hardware, policies
 - Software systems typically
 - EXAMPLES – in the next slides ...

Software Intensive Systems: Examples ... 1

- Example 1: ERP Systems
 - Enterprise Resource Planning systems integrate:
 - Finance
 - HR; – Inventory; – Supply chain; – Compliance;
 - Characteristics:
 - Large codebases
 - Strong consistency requirements
 - Complex workflows
 - Human-in-the-loop interactions

Software Intensive Systems: Examples ... 2

- Example 2: Hospital Information Management Systems (HIMS)
 - Here, software errors can directly impact patient safety.
 - Additional complexity:
 - Medical devices; • Regulatory constraints; • Privacy laws; • Real-time decision support;

Software Intensive Systems: Examples ... 3

- Example 3: University Administration Systems

- Functional modules:

- Admissions
- Academics
- Finance
- Examinations

These systems highlight:

1. Workflow correctness
2. Access control
3. Auditability

At this level, verification must address?

- i. **process correctness**,
- ii. **data integrity**, and
- iii. **policy compliance** — not just code.

After Software Intensive Systems... “Embedded Systems”

- ... we cross a crucial boundary
 - **Embedded system:** a computing system embedded within a larger physical system, designed to perform **dedicated functions**, often under **real-time constraints**.
 - Has a digital controller running a software
 - By reading: Sensors
 - By actuating: Actuators

Examples:

Washing machines

Automotive ECUs

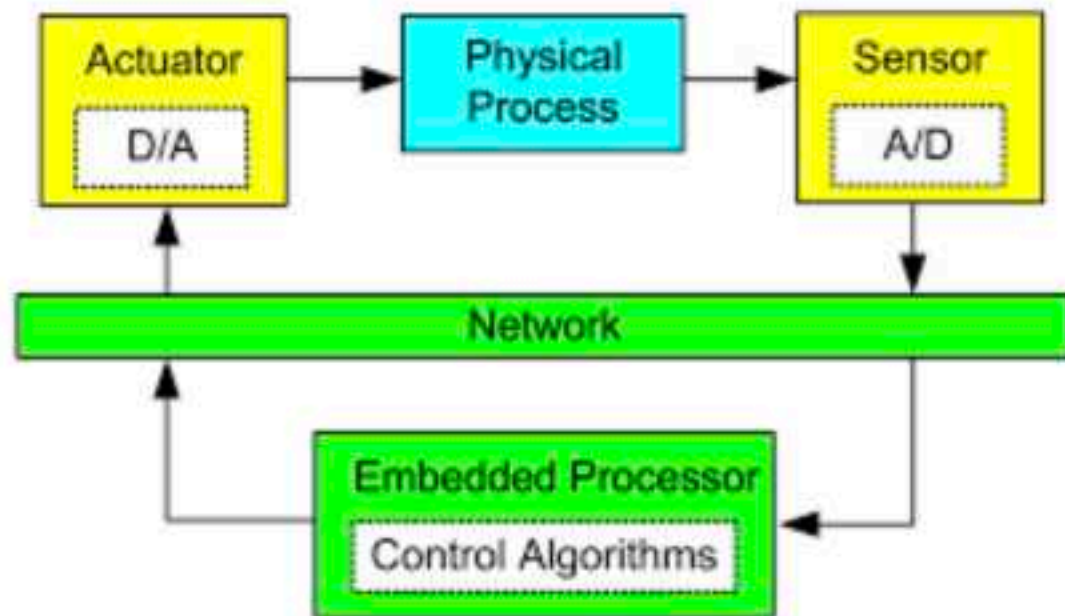
Pacemakers

Flight controllers

- Key Distinction: Pure software systems
 1. Embedded systems **interact with the physical world** Inputs and outputs are **analog**
 2. Time becomes a **first-class constraint**

Embedded System: An Architectural View

- Draw and illustrate



Embedded System: they are Everywhere ... Pervasive ... Ubiquitous



Industrial Robots



GPS Receivers



Digital Cameras



DVD Players

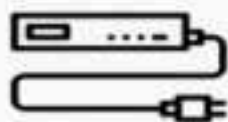


Wireless Routers

Embedded Systems



MP3 Players



Set top Boxes



Gaming Consoles

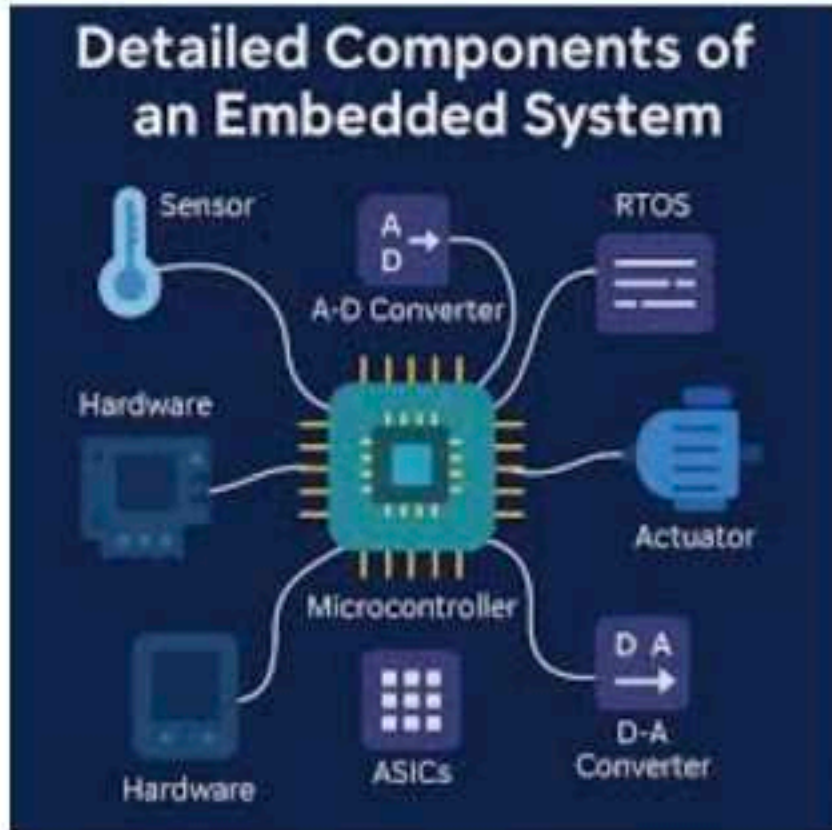


Photocopiers



Microwave Ovens

Embedded System: its composition



Critical Components

- **Sensors**: measure physical quantities
 - **ADC (A2D)**: converts continuous signals into digital values
 - **Processor**: executes control software
 - **DAC (D2A)**: converts digital outputs to analog signals
 - **Actuators**: affect the physical world
- Verification must now consider quantization, timing, and physical effects.

From Embedded Systems to Cyber-Physical Systems (CPS)

- Cyber-Physical System (CPS) tightly integrates:
 - Computation (cyber)
 - Physical processes (continuous/infinite states)
 - Communication

What changes compared to classical embedded systems?

1. Feedback loops are **explicit**
2. Physical dynamics matter
3. Software decisions influence continuous behavior

From Embedded Systems to Cyber-Physical Systems (CPS)

- Cyber-Physical System (CPS) tightly integrates:
 - Computation (cyber)
 - Physical processes (continuous/infinite states)
 - Communication

Examples:

- Autonomous vehicles
- Smart grids
- Medical infusion systems
- Industrial automation

Cyber-Physical Systems (CPS): A verification perspective

- CPS introduces hybrid behavior:
- Discrete software states
- Continuous physical dynamics Verification now must reason about:
 - Timing
 - Hybrid state spaces

WSN: A verification perspective

- Illustrate

Characteristics: Large number of nodes

Limited energy

Wireless communication

Often unattended deployment

- Architecture: Sensor Node $\leftrightarrow$ Wireless Link $\leftrightarrow$ Sink/Gateway

Internet of Things (IoT): A verification perspective

- IoT generalizes by adding:
 - Internet connectivity
 - Cloud backends
 - Mobile and web interfaces
 - Human interaction
- Key verification challenges:
 - End-to-end correctness
 - Security and privacy
 - Responsibility and accountability
 - Certification across organizational boundaries